

**Speak Like A Mathematician:**

included angle:

included side:

hypotenuse:

leg:

congruence criterion:

**The Five Triangle Congruence Criteria**

**SSS:** if three \_\_\_\_\_ of one triangle are congruent to three sides of another, the triangles are congruent.

**SAS:** two sides and the \_\_\_\_\_ angle.      **ASA:** two angles and the \_\_\_\_\_ side.

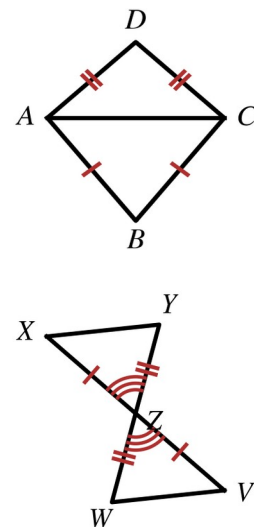
**AAS:** two angles and a \_\_\_\_\_ side.      **HL:** in RIGHT triangles, the \_\_\_\_\_ and one \_\_\_\_\_.

**There is no SSA and no AAA. Same angles can mean a bigger copy; SSA can leave \_\_\_\_\_ different triangles.**

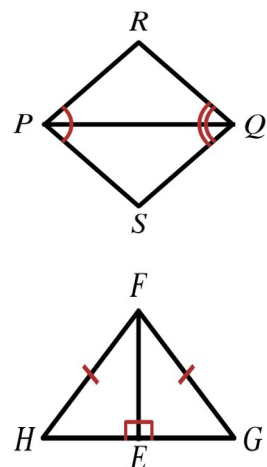
**Example 1 - Name the Criterion**

Each pair of triangles at the right is marked. Name the criterion that proves them congruent.

- a)  $\triangle ABD$  and  $\triangle CBD$
- b)  $\triangle XYZ$  and  $\triangle VWZ$
- c)  $\triangle PQR$  and  $\triangle PQS$
- d)  $\triangle EFG$  and  $\triangle EFH$



SC Honors Geometry - Nappier  
Triangle Rigidity — SSS, SAS, ASA, AAS, HL



## Example 2 - Why SSA and AAA Are Not Criteria

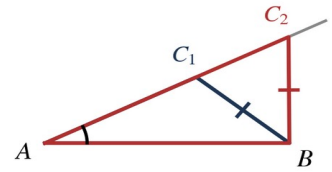
Study the two counterexamples at the right.

- In the first figure, which three parts are congruent in both triangles? Why is that not enough?
- In the second figure, all three angles are congruent. Why are the triangles not congruent?
- Which criteria DO use two sides and an angle? How are they different from SSA?

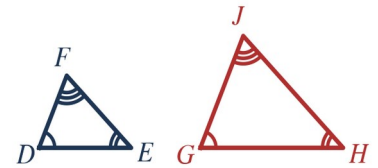
## Example 3 - Find the Value of x

In the diagram,  $\triangle MNO \cong \triangle PQR$ .

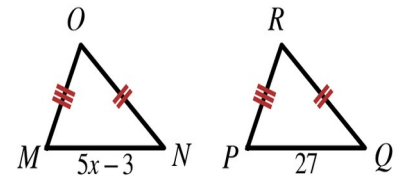
- Which side corresponds to  $\overline{MN}$ ?
- Write and solve an equation for x.
- Check your answer.



same two sides and the same non-included angle —  
but two different triangles



all three angles congruent — but not the same size

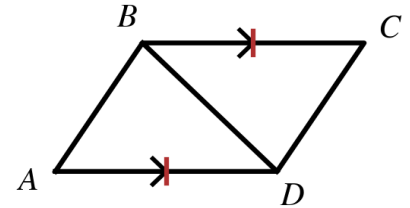


### Example 4 - Proof Using SAS

**Given:**  $\overline{BC} \parallel \overline{AD}$ ,  $\overline{BC} \cong \overline{AD}$

**Prove:**  $\triangle ABD \cong \triangle CDB$

| Statements                                 | Reasons  |
|--------------------------------------------|----------|
| 1. $\overline{BC} \parallel \overline{AD}$ | 1. Given |
| 2.                                         | 2.       |
| 3.                                         | 3.       |
| 4.                                         | 4.       |
| 5.                                         | 5.       |

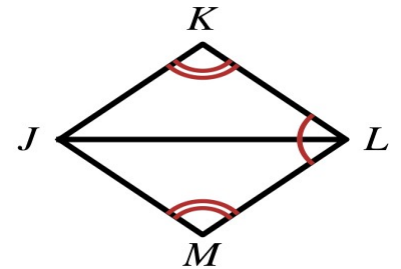


### Example 5 - Proof Using AAS

**Given:**  $\overline{JL}$  bisects  $\angle KLM$ ;  $\angle K \cong \angle M$

**Prove:**  $\triangle JKL \cong \triangle JML$

| Statements                              | Reasons  |
|-----------------------------------------|----------|
| 1. $\overline{JL}$ bisects $\angle KLM$ | 1. Given |
| 2.                                      | 2.       |
| 3.                                      | 3.       |
| 4.                                      | 4.       |
| 5.                                      | 5.       |

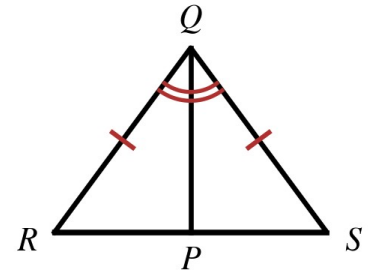


### Example 6 - Proof Using SAS Again

**Given:**  $\overline{QP}$  bisects  $\angle RQS$ ;  $\overline{QR} \cong \overline{QS}$

**Prove:**  $\triangle RQP \cong \triangle SQP$

| Statements                              | Reasons  |
|-----------------------------------------|----------|
| 1. $\overline{QP}$ bisects $\angle RQS$ | 1. Given |
| 2.                                      | 2.       |
| 3.                                      | 3.       |
| 4.                                      | 4.       |
| 5.                                      | 5.       |



### Example 7 - Choose a Criterion

Decide whether the given information is enough to prove the triangles congruent. If it is, name the criterion.

- a) Two pairs of sides and the pair of angles between them.
- b) Two pairs of angles and the pair of sides between them.
- c) Two pairs of angles and a pair of sides NOT between them.
- d) Two pairs of sides and a pair of angles NOT between them.
- e) All three pairs of angles.